

Contact

E-mail: benchugg@cmu.edu

Homepage: <https://benchugg.com/>

Coordinates: GHC 8109, Machine Learning Department, Carnegie Mellon University

Education

2022–	Ph.D. in Machine Learning and Statistics, Carnegie Mellon University Advisor: Aaditya Ramdas Thesis: <i>E-statistics: Advances in concentration, decision-making, and philosophy</i>
2022–2024	MS in Machine Learning, Carnegie Mellon University Advisor: Aaditya Ramdas
2018–2019	M.Sc. in Mathematics (MFoCS), Oxford University Distinction. Advisor: Renaud Lambiotte Thesis: <i>The Graph-Simplex Correspondence and its Algorithmic Foundations</i>
2014–2018	B.Sc. in Mathematics and Computer Science, University of British Columbia Combined Honours with Distinction. Thesis advisor: William Evans. Thesis: <i>A Model for Computing in Dynamic, Resource-Limited Environments</i>

Positions

Summer 2026	Research Scientist Intern, Meta, London, UK
Summer 2024	Visiting Researcher, Centrum Wiskunde & Informatica (CWI), Amsterdam, NL
2021–2022	Lead Research Analyst, RegLab, Stanford Law School, Palo Alto, USA
2019–2021	Research Fellow, RegLab, Stanford Law School, Palo Alto, USA
Summer 2018	Research Intern, RIKEN Center for Advanced Intelligence Project, Tokyo, Japan
2016–2017	NSERC Research Assistant, UBC Algorithms Lab, Vancouver, Canada

Publications

Published or accepted journal papers

- (j10) A variational approach to dimension-free self-normalized concentration. Ben Chugg, Aaditya Ramdas. *Transactions on Machine Learning Research*, 2026
- (j9) E-values as statistical evidence: A comparison to Bayes factors, likelihoods, and p-values. Ben Chugg, Aaditya Ramdas, Peter Grünwald. *Synthese*, 208(3), Art. 130, 2026
- (j8) Nonasymptotic heavy-tailed mean estimation in Banach spaces. Justin Whitehouse, Ben Chugg, Diego Martinez-Taboada, Aaditya Ramdas. *Stochastic Processes and their Applications*, 198, Art. 104959, 2026
- (j7) On admissibility in post-hoc hypothesis testing. Ben Chugg, Tyron Lardy, Aaditya Ramdas, Peter Grünwald. *International Journal of Approximate Reasoning*, 191, Art. 109634, 2026
- (j6) Time-uniform confidence spheres for means of random vectors. Ben Chugg, Hongjian Wang, Aaditya Ramdas. *Transactions on Machine Learning Research* ISSN 2835-8856, 2025
- (j5) A unified recipe for (time-uniform) PAC-Bayes bounds. Ben Chugg, Hongjian Wang, Aaditya Ramdas. *Journal of Machine Learning Research*, 24(372):1–61, 2023

- (j4) Mapping Industrial Poultry Operations at Scale with Deep Learning and Aerial Imagery. Caleb Robinson, Ben Chugg, Brandon Anderson, Juan M. Lavista Ferres, Daniel E. Ho. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 15:7458–7471, 2022
- (j3) Enhancing Environmental Enforcement with Near Real-Time Monitoring: Likelihood-Based Detection of Structural Expansion of Intensive Livestock Farms. Ben Chugg*, Brandon Anderson*, Seiji Eicher, Sandy Lee, Daniel E. Ho. *Journal of Applied Earth Observation and Geoinformation*, 103, Art. 102463, 2021
- (j2) Evaluation of Allocation Schemes of COVID-19 Testing Resources in a Community-Based Door-to-Door Testing Program. Ben Chugg*, Lisa Lu*, Derek Ouyang*, Benjamin Anderson, Raymond Ha, Alexis D’Agostino, Anandi Sujeer, Sarah L. Rudman, Analilia Garcia, Daniel E. Ho. *JAMA Health Forum*, 2(8):e212260, 2021
- (j1) Simultaneous Visibility Representations of Undirected Pairs of Graphs. Ben Chugg, William Evans, Kelvin Wong. *Computational Geometry*, 98, Art. 101788, 2021

Full-length, peer-reviewed conference papers

- (c9) Auditing Fairness by Betting. Ben Chugg, Santiago Cortes-Gomez, Bryan Wilder, Aaditya Ramdas. *Neural Information Processing Systems*, 36:6070–6091, 2023. *Spotlight*
- (c8) Entropy Regularization for Population Estimation. Ben Chugg, Peter Henderson, Jacob Goldin, Daniel E. Ho. *AAAI Conference on Artificial Intelligence*, 37(10):12198–12204, 2023. *Oral*
- (c7) Integrating Reward Maximization and Population Estimation: Sequential Decision-Making for Internal Revenue Service Audit Selection. Peter Henderson, Ben Chugg, Brandon Anderson, Kristen Altenburger, Alex Turk, John Guyton, Jacob Goldin, Daniel E. Ho. *AAAI Conference on Artificial Intelligence*, 37(4):5087–5095, 2023. *Oral*
- (c6) Detecting Environmental Violations with Satellite Imagery in Near Real Time: Land Application Under the Clean Water Act. Ben Chugg*, Nicolas Rothbacher*, Alex Feng, Xiaoqi Long, Daniel E. Ho. *Conference on Information and Knowledge Management*, 3052–3062, 2022
- (c5) Beyond Ads: Sequential Decision-Making Algorithms in Law and Public Policy. Peter Henderson*, Ben Chugg*, Brandon Anderson, Daniel E. Ho. *ACM Symposium on Computer Science & Law*, 87–100, 2022
- (c4) Composable Computation in Leaderless, Discrete Chemical Reaction Networks. Hooman Hashemi, Ben Chugg, Anne Condon. *Intl. Conf. on DNA Computing and Molecular Programming, LIPIcs* 174:3:1–3:18, 2020
- (c3) Simultaneous Visibility Representations of Undirected Pairs of Graphs. Ben Chugg, William Evans, Kelvin Wong. *Canadian Conference on Computational Geometry*, 148–159, 2020
- (c2) Submodular Stochastic Probing with Prices. Ben Chugg, Takanori Maehara. *International Conference on Control, Decision, and Information Technologies*, 60–66, 2019
- (c1) Output-Oblivious Stochastic Chemical Reaction Networks. Ben Chugg, Anne Condon, Hooman Hashemi. *International Conference on Principles of Distributed Systems, LIPIcs* 125:21:1–21:16, 2018

Preprints & Working Papers

- (p7) A/B testing with opportunity costs: Trading off reward and error control. Ben Chugg, Artem Vorozhtsov, Houssam Nassif. 2026
- (p6) Principal-agent hypothesis testing under adaptive experimentation. Ben Chugg, Aritra Ghosh, Aaditya Ramdas. 2026
- (p5) Strategic multi-agent resource allocation via statistical contracts. Ben Chugg, Michael I. Jordan, Aaditya Ramdas. 2026
- (p4) Bentkus-type asymptotic e-values. Diego Martinez-Taboada, Ben Chugg, Aaditya Ramdas. 2026

- (p3) Post-hoc large-sample statistical inference. Ben Chugg*, Etienne Gauthier*, Michael I. Jordan, Aaditya Ramdas, Ian Waudby-Smith. 2026
- (p2) Closed-form empirical Bernstein confidence sequences for scalars and matrices. Ben Chugg, Aaditya Ramdas. 2025
- (p1) Rao-Blackwellized e-variables. Dante de Roos*, Ben Chugg*, Peter Grünwald, Aaditya Ramdas. 2025

Miscellaneous articles

- (m3) Reconciling Risk Reduction and Prevalence Estimation in Public Health Using Batched Bandits, Ben Chugg, Daniel E. Ho. NeurIPS Machine Learning in Public Health, 2021. [Oral](#)
- (m2) The graph-simplex correspondence and its algorithmic foundations. M.Sc. thesis, Oxford, 2019
- (m1) A model for computing in dynamic, resource-limited environments. B.Sc. thesis, UBC, 2018

Selected Awards

2022–2025	NSERC Postgraduate Scholarship–Doctoral (PGS D)
2022–2025	NSERC Graduate Scholarship–Doctoral (CGS D) (Declined)
2018	Mona Leith Memorial Scholarship
2018	Percy Walter Perris Scholarship
2018	Undergraduate Teaching Assistant Award (Computer Science)
2017	Shirley Snelgrove and John Yule Scholarship
2017	NSERC USRA for research in stochastic reaction networks
2016	NSERC USRA for research in graph theory
2014–2018	University of British Columbia Dean’s list
2014	University of British Columbia Chancellor Scholar

Teaching

Teaching Assistant

2025	10-607: Computational foundations for machine learning, CMU
2025	10-606: Mathematical foundations for machine learning, CMU
2024	10-800: Game-theoretic probability, statistics and learning, CMU
2017, 2018	CPSC 420/500: Advanced algorithm design and analysis, UBC
2017	CPSC 320: Intermediate algorithm design and analysis, UBC
2017	CPSC 221: Basic algorithms and data structures, UBC

Talks

2026	E-values as statistical evidence, INFORMS, San Francisco
	E-values as statistical evidence, SAVI, Univ. of Twente
	Strategic multi-agent resource allocation, AI-SDM, Online
2025	On admissibility in post-hoc hypothesis testing, SAVI, Chennai
	A variational approach to concentration, CMU Stat, Pittsburgh
	A variational approach to concentration, UMich, Online
2024	Time-uniform confidence spheres, CWI, Amsterdam
	Auditing by Betting, CMU ML, Pittsburgh
2023	Entropy Regularization for Population Estimation, AAAI, Online

Reviewing

Electronic Journal of Statistics (2023–), NeurIPS (2025, 2026), AAAI (2027), IEEE Transactions on Information Theory (2026–)